

Anaphylaxis with Angioedema



Section I: Scenario Demographics

Scenario Title:	Anaphylaxis with Angioedema		
Date of Development:	(17/01/2017)		
Target Learning Group:	☒ Juniors (PGY 1 - 2)	☐ Seniors (PGY $\geq$ 3)	☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Ahmed Taher
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To review a difficult airway scenario requiring cricothyrotomy
CRM Objectives:	Communicate clearly with team members regarding difficult airway planning
Medical Objectives:	1. Efficiently treats anaphylaxis in an organized manner, including the prioritization of epinephrine administration 2. Recognizes and plans appropriately for the management of a difficult airway 3. Recognizes the need for surgical airway and successfully performs a surgical cricothyrotomy

Case Summary: Brief Summary of Case Progression and Major Events
A 45-year-old patient who has already been seen in the ED begins treatment for pyelonephritis with IV antibiotics. Soon after initiated, she develops stridor and respiratory distress, as part of an anaphylactic reaction. The team is called into the room to assess the patient. After standard anaphylaxis treatment is given, the airway is still of concern. Intubation attempts are not successful and the patient will need a surgical airway.
References
http://lifeinthefastlane.com/cc/anaphylaxis/
https://www.aaaai.org/Aaaai/media/MediaLibrary/PDF%20Documents/Practice%20and%20Parameters/Anaphylaxis-Practice-Parameter-2014.pdf
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1343032/
http://militarymedicine.amsus.org/doi/abs/10.7205/MILMED.172.12.1228

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Section IV: Scenario Script

A. Clinical Vignette: To Read Aloud at Beginning of Case			
You are working a night shift at your local Emergency Department. You are called STAT to the bedside of a patient in the department who was seen by your colleague earlier and has recently been started on IV ceftriaxone for a pyelonephritis. You recall from handover that this is a 45-year-old previously healthy female patient with a diagnosis of a UTI two weeks ago, who returned after failing treatment and was diagnosed with pyelonephritis today. The nurse tells you she started the IV antibiotics and fluids 20 min ago, and then started to experience respiratory distress and a "full body rash".			
B. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin ☐ Mannequin ☐ Standardized Patient ☐ Hybrid ☒ Task Trainer (for Cric)	Realism: <i>Select most important dimension(s)</i>	☒ Conceptual ☒ Physical ☐ Emotional/Experiential ☐ Other: ☐ N/A
Confederates	Brief Description of Role		
Nurse	Assist with delegated tasks, cue learners as to patient's respiratory status and appearance		
C. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☐ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
D. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☒ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☐ Defibrillator	☒ Non-Rebreather Mask	☒ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☒ Other: bougie	
☐ Intraosseous Set-up	☒ LMA	☒ Other: 6-0 ETT	
E. Moulage			
-Urticaria -If mannequin allows: angioedema			
F. Approximate Timing			
Set-Up:	10min	Scenario:	15min
		Debriefing:	15min

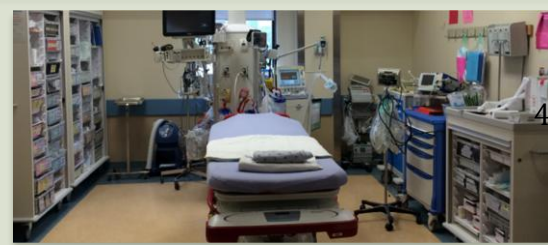
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Section V: Patient Data and Baseline State

A. Patient Profile and History					
Patient Name: Anna Palmer		Age: 45		Weight: 70kg	
Gender: ☐ M ☒ F		Code Status: Full code			
Chief Complaint: Flank Pain, fever					
History of Presenting Illness: Dysuria and treatment for UTI 2 weeks ago. Symptoms worsening with new onset fever, nausea and vomiting. Your colleague in the ED diagnosed her with pyelonephritis, has started IV ceftriaxone, and the general medicine team has been consulted but hasn't been by to assess yet. Twenty minutes after starting the antibiotics, she began to develop stridor, respiratory distress and a pruritic rash.					
Past Medical History:	Healthy	Medications:	Recent nitrofurantoin		
Allergies: None known					
Social History: Lives with partner					
Family History: Non-contributory					
Review of Systems:	CNS:	Feeling Faint			
	HEENT:	Swollen lips, tongue, throat feels "tight"			
	CVS:	No chest pain, but does feel a little dizzy			
	RESP:	Feels hard to breathe			
	GI:	Feels nauseous			
	GU:	Nil			
	MSK:	Nil	INT:	Pruritic rash	
B. Baseline Simulator State and Physical Exam					
☐ No Monitor Display		☐ Monitor On, no data displayed		☒ Monitor on Standard Display	
HR: 130/min	BP: 100/65	RR: 30/min	O ₂ SAT: 88% on room air		
Rhythm: Sinus Tach	T: 37.3°C	Glucose: 8.2mmol/L	GCS: 15 (E4 V5 M6)		
General Status: Notable respiratory distresses with 3-4 word dyspnea					
CNS:	Anxious but alert				
HEENT:	Angioedema (swollen tongue, lips). Clear stridor.				
CVS:	Normal heart sounds, no murmur. Normal peripheral pulses. One peripheral IV in situ.				
RESP:	Stridorous. Also bilateral wheezes and in-drawing				
ABDO:	No tenderness				
GU:	N/A				
MSK:	N/A	SKIN:	Widespread urticarial rash		

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Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: sinus tach HR: 130/min BP: 100 /65 RR: 30/min O ₂ SAT: 88% RA T: 37.3 °C	Patient in severe respiratory distress. Speaking 3-4 word sentences, audible stridor, swollen lips and tongue.	<u>Learner Actions</u> ☐ Apply oxygen and monitors ☐ Obtain 2 nd IV ☐ Initiate IV NS 1L bolus ☐ Discontinue antibiotics ☐ Epinephrine 0.5mg IM ☐ Solumedrol 125mg IV x 1 ☐ Ventolin 5mg nebulized ☐ Consider 5mg nebulized epinephrine ☐ ± Benadryl 50mg iv (low priority) ☐ ± Ranitidine 50 mg iv (low priority)	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - O ₂ placed → Sats to 91% - Epi, ventolin given → HR 150 - If Abx still going at 3 min, RN asks "should we stop the antibiotics?" <u>Triggers</u> <i>For progression to next state</i> - Consideration of intubation → 2. Peri-Intubation - No epi given or no discussion of airway management by 5 min → 3. Desaturating
2. Peri-Intubation HR: 150/min BP: 100/65 RR: 20/min O ₂ SAT: 91%	Respiratory distress, patient getting fatigued – unable to talk, baseline stridor	<u>Learner Actions</u> ☐ Airway assessment ☐ Identify difficult airway ☐ Call for help (extra ERP, anesthesia, ENT) ☐ Prepare intubation equipment, including back-ups (DL, VL, bougie & LMA in room) ☐ Call for surgical airway kit in room, prep equipment, prep neck ☐ Push dose pressor at bedside ☐ Ketamine 1-2mg/kg ± airway ☐ ± Airway topicalization ☐ Consider apneic oxygenation ☐ Consider epinephrine infusion	<u>Modifiers</u> - Propofol used → BP to 70/50 - Paralytic given → unable to bag patient, O ₂ sats drop to 60% <u>Triggers</u> - Intubation attempt → 4. Intubation - Attempt cricothyrotomy → 5. Cricothyrotomy
3. Desaturating HR: 150/min BP: 100/65 RR: 20/min O ₂ SAT: 85%	Same as previous state	<u>Learner Actions</u> ☐ Epinephrine 0.5mg IM ☐ Ventolin 5mg nebulized ☐ Consider 5mg nebulized epinephrine	<u>Modifiers</u> - If no move to airway management 3 min into state → RN to prompt "Are you going to intubate?" <u>Triggers</u> - Consideration of intubation → 2. Peri-Intubation
4. Intubation HR: 150/min BP: 100/65 RR: 20/min shallow O ₂ SAT: 85% (<i>will drop to 70% after first attempt</i>)	Patient deteriorates despite mgmt (Sats drop with any intubation attempt, do not rise with BVM)	<u>Learner Actions</u> ☐ Attempt intubation – (grade 4 view) ☐ Verbalize need for cric	<u>Modifiers</u> - 1 st intubation attempt → "grade IV view," SpO ₂ 70% - BVM after attempt → SpO ₂ 75% - 2 nd attempt → SpO ₂ 60% <u>Triggers</u> - Prepare for cric → 5. Cric



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5. Cricothyrotomy HR: 120 BP: 100/65 RR: 5/ shallow O2 sat: 65%		<u>Learner Actions</u> - ☐ Analgesia/sedation prior to cric (if not yet given) - ☐ Prep the neck - ☐ Vertical skin incision over cricothyroid membrane - ☐ Blunt dissection through subcutaneous tissue - ☐ Horizontal incision through cricothyroid membrane - ☐ Dilate with gloved finger - ☐ Pass bougie into trachea beside finger - ☐ Pass ETT over bougie and intubate trachea	<u>Modifiers</u> - Any method of surgical airway is acceptable <u>Triggers</u> - Cric successful → 6. Resolution
6. Resolution HR: 110 BP: 115/65 RR: 12 O2 sat: 95%	Patient cric +/- sedated	<u>Learner Actions</u> - ☐ Confirm placement (breath sounds, ETCO2, CXR) - ☐ Consult ICU - ☐ ± Epinephrine infusion - ☐ Initiate sedation	<u>END CASE PRN</u>

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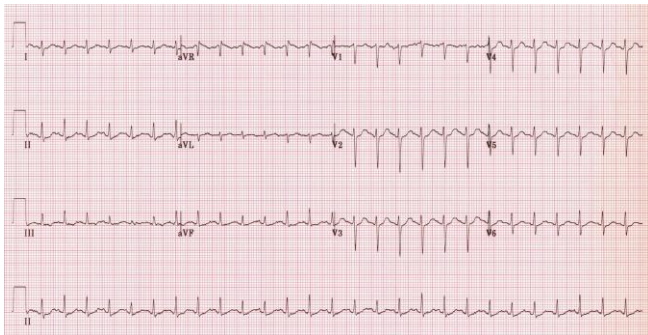


Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results

None provided.

Images (ECGs, CXRs, etc.)



(ECG source:
<http://i0.wp.com/lifeinthefastlane.com/wp-content/uploads/2011/12/sinus-tachycardia.jpg>)



(CXR source:
<https://radiopaedia.org/cases/normal-chest-radiograph-female>)

Ultrasound Video Files (if applicable)

None provided.

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Section VIII: Debriefing Guide

General Debriefing Plan			
☐ Individual	☒ Group	☐ With Video	☒ Without Video
Objectives			
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Medical Objectives:	1. Efficiently treats anaphylaxis in an organized manner, including the prioritization of epinephrine administration 2. Recognizes and plans appropriately for the management of a difficult airway 3. Recognizes the need for surgical airway and successfully performs a surgical cricothyrotomy		
Sample Questions for Debriefing			
1. How did the team initially approach the airway management task? Did the team decide to avoid any particular agents? 2. What equipment should be available at the bedside for an emergency cricothyrotomy? 3. What are the features of difficult cricothyrotomy? Any contraindications to cricothyrotomy? 4. How did you feel in making the decision to perform a cricothyrotomy? Were there any steps you would improve upon thinking about the case now? 5. Was there any hesitation in performing the cricothyrotomy? Why do you think this is? 6. How can a team ensure that a critical procedure, such as a cric, is performed in a timely fashion? Are there any communication tactics that can be used?			
Key Moments			
Recognition of anaphylaxis and initiation of treatment			
Recognition of difficult airway			
Recognition of failed intubation and need for surgical airway			